

LaTeX for PhD Papers — Extended Quick Writing Reference

Paper-writing commands, reusable source blocks, and compiled examples — designed for quick lookup

1. Minimal paper setup

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb,mathtools,bm}
\usepackage{booktabs,threeparttable}
\usepackage{graphicx,subcaption}
\usepackage{natbib}
\usepackage[hidelinks]{hyperref}
\usepackage{microtype}
\usepackage{setspace}

\begin{document}
...
\end{document}
```

Package	What it does in a paper
geometry	margins / page size
amsmath	equations, align, cases
amssymb	extra mathematical symbols
mathtools	extensions to amsmath
bm	bold mathematical symbols
booktabs	professional horizontal table rules
threeparttable	table notes below tables
graphicx	figures and image scaling
subcaption	multi-panel figures
natbib	author–year citations
hyperref	clickable links and references
microtype	improves line breaking / typography
setspace	one-half or double spacing

2. A practical paper preamble

```
\documentclass[11pt]{article}

% Page and typography
\usepackage[margin=1in]{geometry}
\usepackage{microtype}
\usepackage{setspace}

% Math
\usepackage{amsmath,amssymb,mathtools,bm}

% Tables and figures
\usepackage{booktabs,threeparttable}
\usepackage{graphicx,subcaption}

% References and links
\usepackage{natbib}
\usepackage[hidelinks]{hyperref}

% Optional conveniences
\usepackage{enumitem}
\usepackage{xcolor}
```

Good rule: begin with a small preamble. Add packages only when you actually need them.

3. Title, author, affiliation, date

```
\title{Paper Title}
\author{
  First Author\thanks{University A. Email: ...}
  \and
  Second Author\thanks{University B. Email: ...}
}
\date{September 2026}

\begin{document}
\maketitle
```

Common date choices:

```
\date{\today}      % automatic
\date{}            % suppress date
\date{September 2026} % fixed
```

Anonymous draft:

```
\author{Anonymous}
\date{}
```

4. Abstract, keywords, and JEL lines

```
\begin{abstract}
This paper studies ...
\end{abstract}

\medskip
\noindent\textbf{Keywords:} keyword one;
keyword two; keyword three.

\noindent\textbf{JEL Codes:} A10, C10, D10.
```

Compiled style:

Keywords: keyword one; keyword two; keyword three.

JEL Codes: A10, C10, D10.

5. Normal paper section structure

```
\section{Introduction}
\section{Background}
\section{Data}
\section{Method}
\section{Results}
\section{Conclusion}

\subsection{Sample Construction}
\subsubsection{Additional Details}
```

Starred versions suppress numbering:

```
\section*{Acknowledgments}
\subsection*{Data Availability}
```

Compiled hierarchy:

1 Introduction

1.1 Motivation

1.1.1 Additional detail

6. Paragraphs, line breaks, and spacing

Source	Use
blank line	new paragraph
<code>\\</code>	forced line break; use sparingly
<code>\noindent</code>	no paragraph indent
<code>\smallskip</code>	small vertical gap
<code>\medskip</code>	medium vertical gap
<code>\bigskip</code>	large vertical gap
<code>\newpage</code>	immediately start new page
<code>\clearpage</code>	print pending floats, then new page

Draft spacing:

```
\onehalfspacing
\doublespacing
\singlespacing
```

7. Text formatting you actually use

Source	Compiled
<code>\textbf {important}</code>	important
<code>\textit {term}</code>	<i>term</i>
<code>\emph {emphasis}</code>	<i>emphasis</i>
<code>\texttt {file.tex}</code>	<code>file.tex</code>
<code>\underline {text}</code>	<u>text</u>
<code>\textsc {small caps}</code>	SMALL CAPS

Usually prefer `\emph {}` over underlining.

8. Quotes, hyphens, dashes, and special characters

```
‘‘quoted text’’
short-run          % hyphen
1990--2000          % en dash
A result---with a note % em dash
```

Special characters:

`\%` → % `\$` → \$ `\&` → & `\_` → _ `\#` → #

Nonbreaking space:

```
Table~\ref{tab:main}
Figure~\ref{fig:main}
Section~\ref{sec:data}
Author~(2026)
```

The `~` prevents an ugly line break between the word and its number.

9. Footnotes and block quotations

Footnote:

```
The sample is restricted to adults.\footnote{
Additional restrictions are described in Appendix A.}
```

Block quotation:

```
\begin{quote}
A short quotation or indented passage.
\end{quote}
```

Compiled:

A short quotation or indented passage.

10. Labels and cross-references

```
\section{Data}\label{sec:data}

\begin{equation}
a=b+c.
\label{eq:identity}
\end{equation}

\begin{table}[htbp]
\caption{Summary Statistics}
\label{tab:summary}
...
\end{table}

Section~\ref{sec:data}
Equation~\eqref{eq:identity}
Table~\ref{tab:summary}
```

Consistent prefixes save time:

`sec:    eq:    tab:    fig:    app:`

11. Inline math vs. displayed math

Inline math: $x_i+y_i=z_i$ appears in a sentence.

Displayed math:
$$x_i+y_i=z_i.$$

Compiled:

Inline math: $x_i + y_i = z_i$ appears in a sentence.

$$x_i + y_i = z_i.$$

Use displayed math only when the expression deserves visual emphasis.

12. High-frequency math syntax

Source	Result	Source	Result
<code>x_i</code>	x_i	<code>x_{it}</code>	x_{it}
<code>x^2</code>	x^2	<code>x^{t+1}</code>	x^{t+1}
<code>\hat {\theta }</code>	$\hat{\theta}$	<code>\bar {x}</code>	$\bar{x}$
<code>\tilde {x}</code>	$\tilde{x}$	<code>\dot {x}</code>	$\dot{x}$
<code>\frac {a}{b}</code>	$\frac{a}{b}$	<code>\sqrt {x}</code>	$\sqrt{x}$
<code>\sum _{i=1}^n x_i</code>	$\sum_{i=1}^n x_i$	<code>\prod _{i=1}^n x_i</code>	$\prod_{i=1}^n x_i$
<code>\int _a^b f(x)\,dx</code>	$\int_a^b f(x) dx$	<code>\lim _{n\to \infty }</code>	$\lim_{n \rightarrow \infty}$

13. Greek letters and common symbols

Source	Result	Source	Result	Source	Result
<code>\alpha</code>	α	<code>\beta</code>	β	<code>\gamma</code>	γ
<code>\delta</code>	δ	<code>\theta</code>	θ	<code>\lambda</code>	λ
<code>\mu</code>	μ	<code>\sigma</code>	σ	<code>\rho</code>	ρ
<code>\phi</code>	ϕ	<code>\epsilon</code>	ϵ	<code>\omega</code>	ω
<code>\le</code>	$\leq$	<code>\ge</code>	$\geq$	<code>\neq</code>	$\neq$
<code>\approx</code>	$\approx$	<code>\in</code>	$\in$	<code>\notin</code>	$\notin$
<code>\to</code>	$\rightarrow$	<code>\Rrightarrow</code>	$\Rightarrow$	<code>\iff</code>	$\iff$

14. Fonts and operators inside equations

Source	Compiled
<code>\mathbf{x}</code>	x
<code>\bm{\theta}</code>	<i>θ</i>
<code>\mathbb{R}</code>	$\mathbb{R}$
<code>\mathcal{F}</code>	$\mathcal{F}$
<code>\mathrm{Var}(X)</code>	$\mathrm{Var}(X)$
<code>\operatorname{Cov}(X,Y)</code>	$\operatorname{Cov}(X,Y)$
<code>\text{if } x>0</code>	if $x > 0$

Use `\text{...}` for normal prose inside math mode.

15. Align: the default for multi-line equations

<pre>\begin{align} A &= B+C, \label{eq:first}\\ &= D+E, \\ &= F. \label{eq:last} \end{align}</pre>
--

Compiled:

$$A = B + C,$$

$$= D + E,$$

$$= F.$$

(1)

(2)

(3)

Use `align*` when you do not want equation numbers.

16. Long equations: split and aligned

One equation number:

<pre>\begin{equation} \begin{split} A &= {}& B+C+D+E \\ &+F+G+H. \end{split} \end{equation}</pre>

Several aligned lines inside display math:

<pre>\[\begin{aligned} a &= b+c, \\ d &= e+f. \end{aligned} \]</pre>

17. Delimiters, cases, matrices, vectors

Automatic delimiter size:

<pre>\left(\frac{a}{b} \right) \left[x+y \right] \left\{ x:x>0 \right\}</pre>
--

Cases:

<pre>f(x)= \begin{cases} a, & \text{if } x>0, \\ b, & \text{otherwise}. \end{cases}</pre>
--

Matrix:

<pre>\begin{pmatrix} a & b \\ c & d \end{pmatrix}</pre>

Compiled:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad f(x) = \begin{cases} a, & x > 0, \\ b, & \text{otherwise.} \end{cases}$$

18. Math spacing and punctuation

Useful spacing commands:

`\,` small `\;` medium `\quad` large `\!` negative thin space

Example:

<code>\int f(x)\,\mathrm{d}x</code>

produces

$$\int f(x) \, \mathrm{d}x.$$

Equations are part of sentences, so punctuation still matters.

19. Basic paper table

<pre>\begin{table}[htbp] \centering \caption{Summary Statistics} \label{tab:summary} \begin{tabular}{lrrrr} \toprule Variable & Mean & SD & N \\ \midrule Variable A & 1.24 & 0.53 & 1,000 \\ Variable B & 3.10 & 1.22 & 1,000 \\ \bottomrule \end{tabular} \end{table}</pre>

Compiled:

Variable	Mean	SD	N
Variable A	1.24	0.53	1,000
Variable B	3.10	1.22	1,000

Paper default: prefer booktabs; avoid vertical rules.

20. Column alignment and widths

Column letters:

l = left c = center r = right

Fixed-width paragraph column:

```
\begin{tabular}{lp{2.6in}r}
...
\end{tabular}
```

Flexible-width table:

```
\begin{tabularx}{\textwidth}{lXr}
...
\end{tabularx}
```

Suppress outer padding:

```
\begin{tabular}{@{}lrr@{}}
...
\end{tabular}
```

21. Multi-column headers and table panels

```
\begin{tabular}{lcccc}
\toprule
& \multicolumn{2}{c}{Group 1}
& \multicolumn{2}{c}{Group 2} \\
\cmidrule(lr){2-3}\cmidrule(lr){4-5}
Variable & A & B & C & D \\
\midrule
... \\
\bottomrule
\end{tabular}
```

Compiled header:

	Group 1		Group 2	
Variable	A	B	C	D
Example	1	2	3	4

Panel heading:

```
\multicolumn{5}{l}{\textit{Panel A: Main Sample}} \\
```

22. Table notes: threeparttable

```
\begin{table}[htbp]
\centering
\caption{Main Results}
\begin{threeparttable}
\begin{tabular}{lcc}
\toprule
& (1) & (2) \\
\midrule
... \\
\bottomrule
\end{tabular}
\begin{tablenotes}[flushleft]\footnotesize
```

```
\item \textit{Notes:} This table reports ...
\item Standard errors are shown in parentheses.
\end{tablenotes}
\end{threeparttable}
\end{table}
```

This is cleaner than placing a long paragraph manually beneath the table.

23. Decimal-aligned numerical columns

Optional package:

```
\usepackage{siunitx}
```

Then:

```
\begin{tabular}{l S[table-format=2.2]}
\toprule
Variable & {Estimate} \\
\midrule
A & 1.25 \\
B & -0.42 \\
C & 12.30 \\
\bottomrule
\end{tabular}
```

S aligns numbers around the decimal point automatically.

24. Wide tables: scale only when needed

First try smaller text:

```
\begin{table}[htbp]
\centering
\small
...
\end{table}
```

If necessary:

```
\resizebox{\textwidth}{!}{%
\begin{tabular}{lcccccc}
...
\end{tabular}
}
```

Or safer with adjustbox:

```
\begin{adjustbox}{max width=\textwidth}
\begin{tabular}{lcccccc}
...
\end{tabular}
\end{adjustbox}
```

Prefer simplifying the table before shrinking it.

25. Landscape table page

Preamble:

```
\usepackage{pdfscape}
```

Body:

```
\begin{landscape}
\begin{table}[p]
\centering
\caption{Large Table}
...
```

```
\end{table}
\end{landscape}
```

The PDF viewer rotates that page automatically.

26. Import a table generated by R / Stata / Python

If software exports a LaTeX table to `tables/main.tex`:

```
\begin{table}[htbp]
\centering
\caption{Main Results}
\label{tab:main}
\input{tables/main.tex}
\end{table}
```

Keeping generated tables in separate files makes the manuscript much easier to maintain.

27. Basic figure

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.80\textwidth]{figures/main_figure.pdf}
\caption{Main Figure}
\label{fig:main}
\end{figure}
```

Compiled layout idea:



Figure 1: Main Figure

For line charts and plots, vector PDF is usually preferable to screenshots.

28. Figure size, crop, trim, and rotation

Common sizing:

```
\includegraphics[width=\textwidth]{fig.pdf}
\includegraphics[width=0.75\linewidth]{fig.pdf}
```

Crop margins:

```
\includegraphics[
width=0.8\textwidth,
trim={1cm 0.5cm 1cm 0.5cm},
clip
]{fig.pdf}
```

Rotate:

```
\includegraphics[angle=90,width=0.8\textheight]{fig.pdf}
```

Trim order is **left bottom right top**.

29. Two-panel figure

```
\begin{figure}[htbp]
\centering
\begin{subfigure}{0.48\textwidth}
\centering
```

```
\includegraphics[width=\linewidth]{a.pdf}
\caption{Panel A}
\label{fig:a}
\end{subfigure}
\hfill
\begin{subfigure}{0.48\textwidth}
\centering
\includegraphics[width=\linewidth]{b.pdf}
\caption{Panel B}
\label{fig:b}
\end{subfigure}
\caption{Main Figure Title}
\label{fig:panels}
\end{figure}
```

Inside each panel, use `width=\linewidth`.

30. Four-panel figure

```
\begin{figure}[htbp]
\centering
\begin{subfigure}{0.48\textwidth}
\includegraphics[width=\linewidth]{a.pdf}
\caption{A}
\end{subfigure}
\hfill
\begin{subfigure}{0.48\textwidth}
\includegraphics[width=\linewidth]{b.pdf}
\caption{B}
\end{subfigure}

\medskip
\begin{subfigure}{0.48\textwidth}
\includegraphics[width=\linewidth]{c.pdf}
\caption{C}
\end{subfigure}
\hfill
\begin{subfigure}{0.48\textwidth}
\includegraphics[width=\linewidth]{d.pdf}
\caption{D}
\end{subfigure}
\caption{Four-Panel Figure}
\end{figure}
```

31. Figure notes / source notes

Simple method:

```
\begin{figure}[htbp]
\centering
\includegraphics[width=.8\textwidth]{fig.pdf}
\caption{Title}
\label{fig:title}
\begin{minipage}{0.8\textwidth}
\footnotesize
\textit{Notes:} The figure shows ...
\end{minipage}
\end{figure}
```

Use the same width for the note and the figure for a clean appearance.

32. Float placement without fighting LaTeX

Option	Meaning
<code>h</code>	approximately here
<code>t</code>	top of page
<code>b</code>	bottom of page
<code>p</code>	dedicated float page
<code>!</code>	relax placement restrictions
<code>H</code>	exactly here; requires <code>float</code>

Good defaults:

```
\begin{table}[htbp]
\begin{figure}[htbp]
```

To prevent floats from crossing a boundary:

```
\usepackage{placeins}
...
\FloatBarrier
```

33. Citations with natbib

Source	Typical output
<code>\citet{smith2026}</code>	Smith (2026)
<code>\citep{smith2026}</code>	(Smith, 2026)
<code>\citep[p.~5]{smith2026}</code>	(Smith, 2026, p. 5)
<code>\citeauthor{smith2026}</code>	Smith
<code>\citeyear{smith2026}</code>	2026
<code>\citealp{smith2026}</code>	Smith, 2026

Multiple citations:

```
\citep{smith2026,jones2024,lee2023}
```

Text before / after citation:

```
\citep[see][p.~12]{smith2026}
```

34. Citation patterns used in prose

```
\citet{smith2026} shows that ...
```

```
Related work reaches similar conclusions
\citep{jones2024,lee2023}.
```

```
This mechanism is discussed by
\citet[p.~18]{smith2026}.
```

Compiled style:

Smith (2026) shows that ...

Related work reaches similar conclusions (Jones, 2024; Lee, 2023).

35. Bibliography commands

At the end of the paper:

```
\bibliographystyle{plainnat}
\bibliography{references}
```

Your project then contains:

```
main.tex
references.bib
```

Many journals provide their own bibliography style. If so, replace `plainnat` with the required style rather than manually formatting entries.

36. BibTeX: journal article

```
@article{smith2026,
  author = {Smith, Jane and Lee, John},
  title  = {Article Title},
  journal = {Journal Name},
  year   = {2026},
  volume = {10},
  number = {2},
  pages  = {100--130},
  doi    = {10.xxxx/xxxxx}
}
```

Keep capitalization that BibTeX must preserve inside braces:

```
title = {The Effects of {AI} on Productivity}
```

37. BibTeX: book, chapter, working paper

Book:

```
@book{bookkey,
  author = {Last, First},
  title  = {Book Title},
  year   = {2026},
  publisher = {Publisher}
}
```

Chapter:

```
@incollection{chapterkey,
  author = {Last, First},
  title  = {Chapter Title},
  booktitle = {Book Title},
  editor  = {Editor, Name},
  year    = {2026},
  publisher = {Publisher},
  pages   = {1--25}
}
```

Working paper:

```
@techreport{wpkey,
  author = {Last, First},
  title  = {Working Paper Title},
  institution = {NBER},
  year   = {2026},
  number = {12345}
}
```

38. URLs, DOI links, and project links

```
\url{https://example.com}
```

```
\href{https://example.com}{Project website}
```

```
\href{https://doi.org/10.xxxx/xxxx}{DOI}
```

If a URL causes an overfull line, use `\url{...}` rather than typing the URL as ordinary text.

39. Hyperref: PDF metadata

Optional:

```
\hypersetup{
  pdftitle = {Paper Title},
  pdfauthor = {Your Name},
  pdfsubject = {Economics},
  pdfkeywords = {keyword1, keyword2}
}
```

This affects PDF metadata, not the visible title page.

40. Acknowledgments and data statements

```
\section*{Acknowledgments}
I thank ...

\section*{Data Availability}
Replication files are available at ...

\section*{Conflict of Interest}
The authors declare ...
```

Use only the statements required by your journal, department, or data provider.

41. Lists inside a paper

Bullets:

```
\begin{itemize}
\item First contribution.
\item Second contribution.
\end{itemize}
```

Numbered list:

```
\begin{enumerate}
\item First step.
\item Second step.
\end{enumerate}
```

Custom labels:

```
\begin{enumerate}[label=(\alph*)]
\item ...
\item ...
\end{enumerate}
```

42. Theorem / proposition / lemma / assumption

Preamble:

```
\newtheorem{theorem}{Theorem}
\newtheorem{proposition}{Proposition}
\newtheorem{lemma}{Lemma}
\newtheorem{assumption}{Assumption}
```

Body:

```
\begin{proposition}
State the proposition here.
\end{proposition}

\begin{proof}
Write the proof here.
\end{proof}
```

Useful in theory papers and technical appendices.

43. Definitions and remarks with amsthm

```
\theoremstyle{definition}
\newtheorem{definition}{Definition}

\theoremstyle{remark}
\newtheorem*{remark}{Remark}
```

Then:

```
\begin{definition}
A definition goes here.
\end{definition}

\begin{remark}
An unnumbered remark goes here.
\end{remark}
```

44. Appendix basics

```
\appendix

\section{Additional Results}
\label{app:results}

\section{Proofs}
\label{app:proofs}
```

Then reference it as:

```
See Appendix~\ref{app:results}.
```

Place `\appendix` after the bibliography if that matches your preferred manuscript order.

45. Appendix equations, tables, and figures

After `\appendix`, standard section numbering usually becomes A, B, C, etc. For explicit appendix-style float numbering, you can reset counters:

```
\setcounter{table}{0}
\renewcommand{\thetable}{A\arabic{table}}

\setcounter{figure}{0}
\renewcommand{\thefigure}{A\arabic{figure}}
```

For multiple appendix sections, more elaborate numbering may be handled by the journal template.

46. Large projects: separate files

Suggested structure:

```
paper/
main.tex
references.bib
sections/
  intro.tex
  background.tex
  data.tex
  results.tex
tables/
  summary.tex
figures/
  figure1.pdf
```

Inside `main.tex`:

```
\input{sections/intro}
\input{sections/background}
```

```
\input{sections/data}
\input{sections/results}
```

No `\begin{document}` is needed inside the imported section files.

47. input vs. include

Command	Practical use
<code>\input{file}</code>	insert content exactly here; best default for paper sections/tables
<code>\include{file}</code>	starts on a new page and creates auxiliary files; useful for book-like projects

For most research manuscripts, `\input` is simpler.

48. Reusable custom commands

```
\newcommand{\E}{\mathbb{E}}
\newcommand{\R}{\mathbb{R}}
\newcommand{\vect}[1]{\bm{#1}}
\newcommand{\secref}[1]{Section~\ref{#1}}
\newcommand{\figref}[1]{Figure~\ref{#1}}
\newcommand{\tabref}[1]{Table~\ref{#1}}
```

Then:

```
$$E[X]$$
$$X\text{in}\mathbb{R}$$
\vect{x}
\figref{fig:main}
```

Use custom commands for notation or wording you expect to change globally.

49. Draft comments and visible TODOs

Invisible source comment:

```
% Rewrite this paragraph.
% Check citation.
```

Visible draft note:

```
\textcolor{red}{TODO: Add citation.}
```

Reusable command:

```
\newcommand{\todo}[1]{%
\textcolor{red}{[TODO: #1]}}
```

Before submission, search the source for `TODO`, `FIXME`, question marks, and colored notes.

50. Temporarily hide a large block

Without another package:

```
\iffalse
This entire block is ignored by LaTeX.
It can contain multiple paragraphs.
\fi
```

Useful when testing whether a section, figure, or table is causing an error.

51. Page layout and paragraph style

Margins:

```
\usepackage[margin=1in]{geometry}
```

Traditional indented paragraphs:

```
\setlength{\parindent}{1.5em}
```

```
\setlength{\parskip}{0pt}
```

Block-style paragraphs:

```
\setlength{\parindent}{0pt}
\setlength{\parskip}{6pt}
```

Do not combine large paragraph indentation with large paragraph spacing unless a template explicitly requests it.

52. Page numbering and title-page control

Normal page numbers:

```
\pagestyle{plain}
```

Hide only the current page number:

```
\thispagestyle{empty}
```

Restart page numbering:

```
\setcounter{page}{1}
\pagenumbering{arabic}
```

Roman numerals, when needed:

```
\pagenumbering{roman}
```

53. Submission drafts: double spacing

```
\usepackage{setspace}
...
\doublespacing
```

Or only a portion:

```
\begin{spacing}{1.5}
Text here.
\end{spacing}
```

Use the journal or instructor specification if one exists.

54. Submission drafts: line numbers

Optional package:

```
\usepackage{lineno}
```

Body:

```
\linenumbers
```

Turn them off later with:

```
\nolinelnumbers
```

Some journal submission systems request line-numbered manuscripts.

55. Anonymous vs. named versions

A simple manual switch:

```
% Anonymous submission
\author{Anonymous}

% Working-paper version
% \author{Your Name\University}
```

For a long-running project, a Boolean switch or separate title file can reduce mistakes between versions.

56. Keep a figure or table near discussion

Usually let LaTeX float it. If you need a boundary:

```
\usepackage{placeins}
...
End of results discussion.
\FloatBarrier
\section{Robustness}
```

If you instead use `[H]` everywhere, page layout often becomes worse.

57. Text sizes inside tables and figures

From larger to smaller:

`\normalsize` `\small` `\footnotesize` `\scriptsize`

Example:

```
\begin{table}[htbp]
\centering
\small
...
\end{table}
```

Avoid `\tiny` in a main paper unless there is no reasonable alternative.

58. Centering: environment vs. command

For floats:

```
\begin{figure}[htbp]
\centering
...
\end{figure}
```

For standalone material:

```
\begin{center}
Centered text.
\end{center}
```

Inside a float, `\centering` is generally cleaner than nesting a `center` environment.

59. Minipage for side-by-side text or notes

```
\begin{minipage}[t]{0.48\textwidth}
Left block of text.
\end{minipage}
\hfill
\begin{minipage}[t]{0.48\textwidth}
Right block of text.
\end{minipage}
```

Useful for custom title pages, figure notes, or side-by-side nonfloating material.

60. Prevent awkward line breaks

Use nonbreaking spaces:

```
Figure~1
Table~2
Section~3
p.~10
10~percent
```

If a particular phrase should stay together:

```
\mbox{New York}
```

Use `\mbox` sparingly because it disables line breaking inside the box.

61. Common table / figure caption pattern

```
\caption{Short Informative Title}
\label{tab:descriptive-name}
```

Put `\label` immediately after `\caption` for tables and figures. This avoids cross-reference errors.

62. A clean table-note convention

A common pattern is:

```
\textit{Notes:} This table reports ...
Sample restrictions are ...
Standard errors are ...
Significance markers are ...
```

The note should explain the table sufficiently without forcing the reader to search the main text for basic definitions.

63. A clean figure-note convention

```
\textit{Notes:} The figure plots ...
The shaded region indicates ...
The sample includes ...
Source: ...
```

Keep implementation detail out of the caption when it belongs in the text or appendix.

64. Referencing several items in one sentence

```
Tables~\ref{tab:a} and~\ref{tab:b}
```

```
Figures~\ref{fig:a}--\ref{fig:c}
```

```
Sections~\ref{sec:data} and~\ref{sec:results}
```

The double hyphen `--` renders an en dash for a range.

65. Equation references inside prose

```
Equation~\eqref{eq:main} defines the object.
```

```
Using \eqref{eq:first} and \eqref{eq:second},
we obtain ...
```

Use `\eqref` rather than manually typing parentheses around `\ref`.

66. Text inside equations

```
\[
f(x)=0 \quad \text{for all } x\in\mathcal{X}.
\]
```

Compiled:

$$f(x) = 0 \quad \text{for all } x \in \mathcal{X}.$$

Do not write ordinary words as bare math letters.

67. Underbrace / overbrace for explanation

```
\[
A = \underbrace{B+C}_{\text{first component}}
+ \underbrace{D}_{\text{second component}}.
\]
```

Compiled:

$$A = \underbrace{B+C}_{\text{first component}} + \underbrace{D}_{\text{second component}}.$$

Useful in exposition, but avoid over-annotating final journal equations.

68. Reserved characters — memorize these

These characters have a special meaning in LaTeX source. To print them literally, use the form in the second column.

Character	Type this	Why it is special
#	\#	macro parameter character
\$	\\$	starts / ends inline math
%	\%	starts a comment; rest of source line is ignored
&	\&	column / alignment separator
-	_	subscript in math mode
{	\{	begins a group / command argument
}	\}	ends a group / command argument
~	\textasciitilde	~ itself means a nonbreaking space
^	\textasciicircum	^ is superscript syntax in math
\	\textbackslash	begins a LaTeX command

Most common paper examples:

```
10\% of the sample
R\&D expenditure
The fee is \$500.
data\clean.csv
Figure \#2
\{A,B,C\}
```

Dangerous one: an unescaped percent sign comments out everything to the right of it in the source, often causing mysterious missing text or a later brace error.

69. Literal code, filenames, URLs, and special text

For a short literal string, \verb is often easier than escaping every character:

```
\verb|data_final_v2.csv|
\verb|beta_hat = x_1/x_2|
```

For URLs and paths:

```
\url{https://example.com/a_b?x=1&y=2}
\path{tables/main_results_v3.tex}
```

These commands handle many special characters for you. Do **not** use \verb inside section titles, captions, footnotes, or other command arguments.

For literal angle brackets in prose, use:

```
\textless value \textgreater
```

70. Quotes, dashes, ellipses, and spaces

Want	Type	Compiled idea
English quotes	“‘quoted text’”	“quoted text”
Hyphen	high-frequency	high-frequency
Number range	1990--2000	1990–2000
Em dash	result---however	result—however
Ellipsis	\ldots	...
No line break	Table~\ref{tab:main}	keep label with number
Forced normal space	Dr.\ Smith	prevents sentence-sized spacing

Useful habits in papers:

```
Figure~\ref{fig:event}
Table~\ref{tab:main}
Section~\ref{sec:data}
Appendix~\ref{app:proofs}
Dr.\ Smith
U.S.\ data
```

Plain LaTeX source using “ ‘ ’ ”, --, and --- is usually more portable than pasted typographic quote or dash characters.

71. Common writing mistakes in LaTeX papers

Avoid	Prefer
\\ after every paragraph	blank line between paragraphs
repeated line breaks for vertical space	sectioning or controlled spacing
\$\$... \$\$	\[... \] or equation
\$for all x\$	\text{for all } x \text{ inside math}
manual “Figure 3”	Figure~\ref{fig:key}
manual “Table 2”	Table~\ref{tab:key}
label before figure/table caption	\caption then \label
shrinking a whole table immediately	fix columns / headings first
using [H] for every float	start with [htbp]
spaces to align text	table, align, or proper environments
manual bibliography entries for a large paper	BibTeX / BibLaTeX database

Paragraph rule: a double backslash means “line break,” not “new paragraph.” A blank source line starts a new paragraph.

72. Common equation-writing mistakes

An equation is grammatically part of the sentence. Punctuate it accordingly.

Better:

```
The objective can be written as
\begin{equation}
Q(\theta)=\sum_{i=1}^n q_i(\theta),
\label{eq:objective}
\end{equation}
where $\theta$ denotes the parameter vector.
```

Common mistakes:

- Use \text {...} for ordinary words inside math.
- Use built-in operators such as \log , \exp , \sin , and \max ; do not type them as ordinary italic letters.
- Do not mix unmatched \$, \langle, \rangle, \lceil, or \rceil delimiters.
- Avoid putting long prose inside display math just to center it.

- Put `\label` inside the numbered equation environment you want to reference.

73. Common compile sequence with BibTeX

Classic workflow:

```
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

Why multiple passes? LaTeX needs later passes to resolve citations, equation numbers, figure numbers, table numbers, and page references.

74. Easier compiling with latexmk

If available:

```
latexmk -pdf main.tex
```

Clean auxiliary files:

```
latexmk -c
```

Full cleanup including PDF:

```
latexmk -C
```

Editors such as Overleaf and VS Code + LaTeX Workshop usually automate the repeated passes.

75. Fast debugging: common errors

Symptom	Check first
Undefined control sequence	typo, wrong command name, or missing package
Missing \$ inserted	math command / underscore / caret outside math mode
Extra alignment tab &	unescaped & or too many table/alignment columns
Missing } inserted	unmatched brace, often earlier than the reported line
Runaway argument?	missing closing brace or environment end
Environment mismatch	wrong / missing <code>\end {...}</code>
Reference shows ??	compile again; verify unique label and <code>\ref</code> key
Citation shows ?	wrong citation key, missing <code>.bib</code> , or BibTeX pass
Multiply-defined label	same <code>\label</code> used more than once
File ...sty not found	package not installed / wrong package name
File ... not found	image/input path, capitalization, or extension
Overfull \hbox	long URL, equation, table, or unbreakable text
Underfull \hbox	awkward line stretching; often only a warning
Float(s) lost / misplaced	float nested illegally or placement too restrictive

Best debugging rule: fix the *first* reported LaTeX error first. Later errors are often consequences of that first one.

If an error seems unrelated to the reported line: inspect several lines *above* it for an unescaped percent sign, unmatched \$, missing brace, or missing `\end {...}`.

76. Overfull lines: safe fixes

For URLs:

```
\url{https://very-long-address.example/...}
```

For equations: use `align`, `split`, or `multline` rather than shrinking the font.

For tables: shorten headings, use `tabularx`, reduce columns, or move a very wide table to landscape.

Last-resort local relaxation:

```
\begin{sloppypar}
A difficult paragraph ...
\end{sloppypar}
```

77. Figure / table appears too late

Try, in this order:

1. use `[htbp]`;
2. reduce the float's size;
3. move its source slightly earlier;
4. use `\FloatBarrier` at a logical boundary;
5. use `\clearpage` before an appendix or references if appropriate.

Avoid solving every float problem with `[H]`.

78. Full copy-ready paper skeleton

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb,mathtools,bm}
\usepackage{booktabs,threeparttable}
\usepackage{graphicx,subcaption}
\usepackage{natbib}
\usepackage[hideinks]{hyperref}
\usepackage{microtype}
\usepackage{setspace}
```

```
\title{Paper Title}
\author{Your Name\University Name}
\date{\today}
```

```
\begin{document}
\maketitle
```

```
\begin{abstract}
One-paragraph abstract.
\end{abstract}
```

```
\noindent\textbf{Keywords:} ...
```

```
\noindent\textbf{JEL Codes:} ...
```

```
\section{Introduction}\label{sec:intro}
\input{sections/intro}
```

```
\section{Background}\label{sec:background}
\input{sections/background}
```

```
\section{Data}\label{sec:data}
\input{sections/data}
```

```
\section{Method}\label{sec:method}
\input{sections/method}
```

```
\section{Results}\label{sec:results}
\input{sections/results}
```

```
\section{Conclusion}\label{sec:conclusion}
\input{sections/conclusion}
```

```
\bibliographystyle{plainnat}
\bibliography{references}
```

```
\appendix
\section{Additional Material}
\input{sections/appendix}

\end{document}
```

79. Useful optional packages by task

Task	Package
Landscape pages	pdflscape
Float barriers	placeins
Exact float placement	float
Decimal alignment	siunitx
Fit wide object to page	adjustbox
Multiple table rows	multirow
Better lists	enumitem
Colored draft notes	xcolor
Line numbers	lineno
Subfigures	subcaption

Do not load optional packages unless you need the feature.

80. Commands to recognize instantly

```
\documentclass \usepackage
\begin \end
\title \author
\date \maketitle
\begin{abstract} \section
\subsection \textbf
\textit \emph
\footnote \label
\ref \eqref
$...$ \[...\]
~
\frac \sqrt
\sum \int
\hat \bar
```

```
\mathbb \bm
\left \right
\text \operatorname
\begin{align} \begin{cases}
\begin{table} \toprule
\midrule \bottomrule
\multicolumn \cmidrule
\caption \includegraphics
\begin{figure} \citet
\citep \bibliography
\input \appendix
\newcommand \url
\href \clearpage
\FloatBarrier \resizebox
```

81. What to memorize vs. look up

Memorize: sections, text emphasis, math mode, subscripts/superscripts, `frac`, `align`, `label/ref`, basic tables, basic figures, `caption`, `citet/citep`, `input`, and `appendix`.

Look up when needed: complex multi-panel tables, unusual symbols, custom theorem numbering, journal-specific classes, landscape material, long tables, TikZ, bibliography-style edge cases, and specialized submission formatting.

Fast workflow: copy a working block, change one thing, compile, repeat.

82. Final paper-writing sanity check

Before sending a draft, check:

- every figure and table is referenced in the text;
- captions and notes explain what the reader needs;
- no references or citations show ?? or ?;
- no visible TODO / FIXME notes remain;
- figure fonts are readable at 100% PDF zoom;
- tables do not run outside margins;
- bibliography entries are complete and consistent;
- the anonymous version really removes identifying information;
- appendix labels and references work;
- the PDF opens cleanly after a full compile from scratch.